
MA 607: Numerical Analysis

Course Code	MA 607
Course Name	Numerical Analysis
L-T-P-C	2-0-2-3
Prerequisites	Basic Knowledge in FORTRAN/C/C++/Matlab/Scilab
Intended for	M.A./Ph.D., B.Tech. 3rd and 4th year students.
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- Approximate Numbers and Significant Digits, Propagation of errors, Different types of errors, Backward error analysis, Sensitivity and conditioning, Stability and accuracy.
- Nonlinear equations, Bisection method, Newton's method and its variants, Fixed point iterations, Convergence analysis.
- Finite differences, Polynomial interpolation, Hermite interpolation, Spline interpolation. Numerical integration, Trapezoidal and Simpson's rules, Newton-Cotes formula, Gaussian quadrature.
- **Initial Value Problem (IVP):** Taylor series method, Euler and modified Euler methods, Runge-Kutta methods, Multistep methods, Predictor-Corrector method.
- **Boundary Value Problem (BVP):** Solution of Boundary Value Problem by Finite Difference Method.

Text Books

1. K. E. Atkinson, **An Introduction to Numerical Analysis**, 2nd Edition, John Wiley, 2008.
2. S. D. Conte and Carl de Boor, **Elementary Numerical Analysis** McGraw Hill, 1988.

References

1. M. T. Heath, **Scientific Computing: An Introductory Survey**, McGraw Hill, 2002.
2. A. Ralston and P. Rabinowitz, **A First Course in Numerical Analysis**, Dover Publications, 2001.